



Constructing a Layer 3 Off-Path WLAN

Difficulty (out of 5 stars): ★★★★★

Recommended study time: 3 hours

Contents

1 About This Lab	2
1.1 Task Description	2
1.2 Objectives	2
2 Task Preparation	2
2.1 Network Topology	2
2.2 Initial Configuration	3
3 Task Implementation	3
3.1 Lab Planning	3
3.2 Configuring Network Connectivity	5
3.3 Configuring AP Onboarding	8
3.4 Configuring WLAN Services	11
4 Verification	13
4.1 Verifying that APs Properly Broadcast Signals	13
4.2 Verifying STA Connections and Network Connectivity	14
5 Quiz	15



1 About This Lab

1.1 Task Description

In Layer 3 WLAN networking, the off-path networking mode is an overlay to the existing network, requiring little reconstruction and facilitating deployment. In addition, the direct or tunnel forwarding mode can be used according to control requirements for STAs.

Upon completing this task, you will gain a clear understanding of Layer 3 WLAN networking in off-path mode and be able to configure Layer 3 off-path WLAN networking.

1.2 Objectives

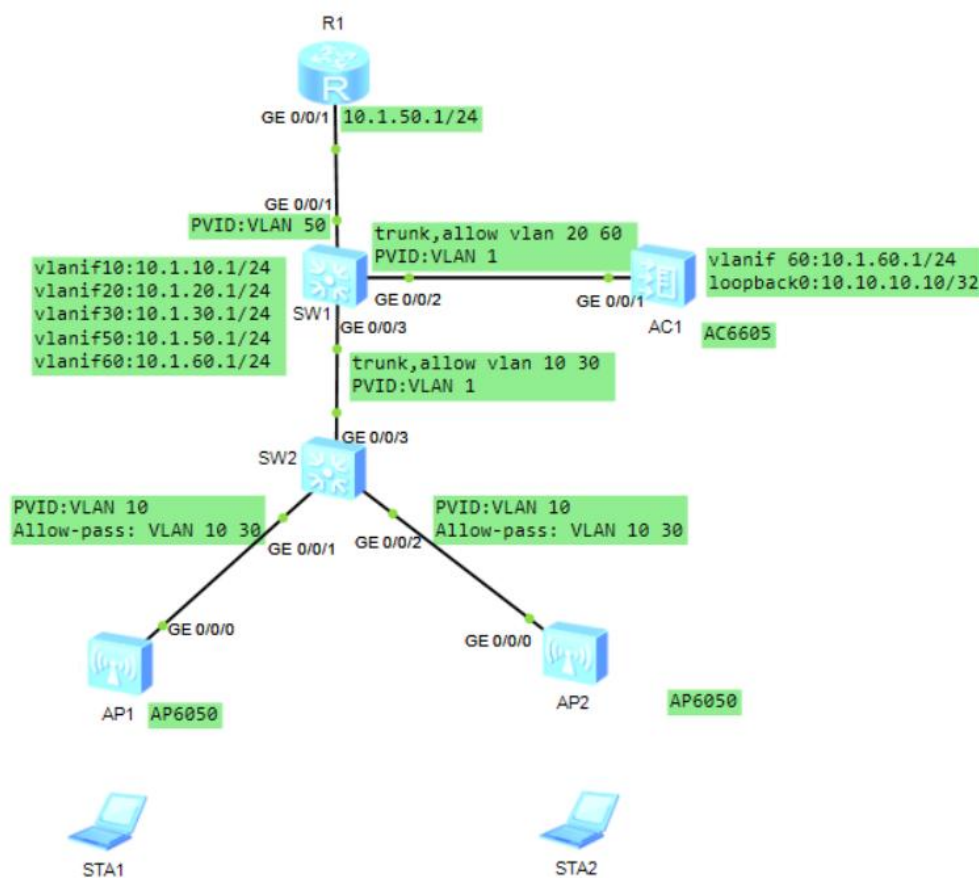
On completing this task, you will be able to:

1. Understand the Layer 3 networking mode.
2. Understand the advantages of off-path networking.
3. Understand how to configure WLAN services.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Layer 3 off-path WLAN networking topology





2.2 Initial Configuration

- Initial configuration of SW1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname SW1
[SW1]
```

- Initial configuration of SW2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname SW2
[SW2]
```

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]
```

- Initial configuration of the AC

```
<AC6605>system-view
Enter system view, return user view with Ctrl+Z.
[AC6605]sysname AC
[AC]
```

- Initial configuration of AP1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]ap-sysname AP2

Info: The configuration takes effect after the AP is restarted.

[Huawei]
# The restart takes effect on the physical device but not on eNSP.
```

- Initial configuration of AP2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]ap-sysname AP2

Info: The configuration takes effect after the AP is restarted.

[Huawei]
# The restart takes effect on the physical device but not on eNSP.
```

3 Task Implementation

3.1 Lab Planning

In this lab, Layer 3 off-path networking is used. SW1 is used as the gateway for both APs and STAs to provide network services for enterprise employees and guests.

**Table 3-1** VLAN interface types and parameters

Device	Interface	Interface Type	VLAN Parameter
SW1	GE0/0/1	Access	PVID: VLAN 50
	GE0/0/2	Trunk	PVID: 1 Allow-pass: VLANs 20 and 60
	GE0/0/3	Trunk	PVID: 1 Allow-pass: VLANs 10 and 30
SW2	GE0/0/1	Trunk	PVID:VLAN 10 Allow-pass: VLANs 10 and 30
	GE0/0/2	Trunk	PVID:VLAN 10 Allow-pass: VLANs 10 and 30
	GE0/0/3	Trunk	PVID: 1 Allow-pass: VLANs 10 and 30
AC	GE0/0/1	Trunk	PVID: 1 Allow-pass: VLANs 20 and 60

Table 3-2 IP address planning

Device	Interface	IP Address
R1	GE0/0/1	10.1.50.1/24
SW1	VLANIF 10	10.1.10.1/24
	VLANIF 20	10.1.20.1/24
	VLANIF 30	10.1.30.2/24
	VLANIF 50	10.1.50.2/24
	VLANIF 60	10.1.60.2/24
AC	VLANIF 60	10.1.60.1/24
	Loopback 0	10.10.10.10/32

Table 3-3 WLAN data planning

Data	Configuration
DHCP server	SW1 functions as a DHCP server to assign IP addresses to APs and STAs and as the gateway for APs and STAs.
IP address pool for APs	10.1.10.2–10.1.10.254/24



Data	Configuration	
IP address pool for STAs	Employee: 10.1.20.2–10.1.20.254/24 Guest: 10.1.30.2–10.1.30.254/24	
AC source interface	Loopback 0	
AP group	Name: Huawei Referenced profiles: VAP profile and regulatory domain profile	
Regulatory domain profile	Name: Huawei Country code: CN	
SSID profiles	Name: Employee SSID name: Employee	Name: Guest SSID name: Guest
Security profiles	Name: Employee Security policy: WPA2+PSK+AES Password: a1234567	Name: Guest Security policy: open system
VAP profiles	Name: Employee Forwarding mode: tunnel forwarding Service VLAN: VLAN 20 Referenced profiles: SSID profile and security profile	Name: Guest Forwarding mode: direct forwarding Service VLAN: VLAN 30 Referenced profiles: SSID profile and security profile

3.2 Configuring Network Connectivity

Configure the corresponding VLANs and interface types on the devices to implement Layer 2 transparent transmission of services.

1. Create VLANs on SW1.

Create VLANs 10, 20, 30, 50, and 60 on SW1.

```
[SW1]vlan batch 10 20 30 50 60
```

Info: This operation may take a few seconds. Please wait for a moment...done.

```
[SW1]
```

2. Configure interfaces on SW1.

On SW1, configure the interface types according to Table 3-1 and allow packets in the corresponding VLANs to pass through.

```
[SW1]interface GigabitEthernet 0/0/1
```

```
[SW1-GigabitEthernet0/0/1]port link-type access
```

```
[SW1-GigabitEthernet0/0/1]port default vlan 50
```

```
[SW1-GigabitEthernet0/0/1]quit
```

```
[SW1]interface GigabitEthernet 0/0/2
```

```
[SW1-GigabitEthernet0/0/2]port link-type trunk
```

```
[SW1-GigabitEthernet0/0/2]port trunk allow-pass vlan 20 60
```



```
[SW1-GigabitEthernet0/0/2]quit
[SW1]interface GigabitEthernet 0/0/3
[SW1-GigabitEthernet0/0/3]port link-type trunk
[SW1-GigabitEthernet0/0/3]port trunk allow-pass vlan 10 30
[SW1-GigabitEthernet0/0/3]quit
[SW1]
```

3. Create VLANs on SW2.

Create VLANs 10 and 30 on SW2.

```
[SW2]vlan batch 10 30
Info: This operation may take a few seconds. Please wait for a moment...done.
[SW2]
```

4. Configure interfaces on SW2.

On SW2, configure the interface types according to Table 3-1 and allow packets in the corresponding VLANs to pass through.

```
[SW2]interface GigabitEthernet 0/0/1
[SW2-GigabitEthernet0/0/1]port link-type trunk
[SW2-GigabitEthernet0/0/1]port trunk pvid vlan 10
[SW2-GigabitEthernet0/0/1]port trunk allow-pass vlan 10 30
[SW2-GigabitEthernet0/0/1]quit
[SW2]interface GigabitEthernet 0/0/2
[SW2-GigabitEthernet0/0/2]port link-type trunk
[SW2-GigabitEthernet0/0/2]port trunk pvid vlan 10
[SW2-GigabitEthernet0/0/2]port trunk allow-pass vlan 10 30
[SW2-GigabitEthernet0/0/2]quit
[SW2]interface GigabitEthernet 0/0/3
[SW2-GigabitEthernet0/0/3]port link-type trunk
[SW2-GigabitEthernet0/0/3]port trunk allow-pass vlan 10 30
[SW2-GigabitEthernet0/0/3]quit
[SW2]
```

5. Create VLANs on the AC.

Create VLANs 20 and 60 on the AC.

```
[AC]vlan batch 20 60
Info: This operation may take a few seconds. Please wait for a moment...done.
[AC]
```

6. Configure interfaces on the AC.

On the AC, configure the interface types according to Table 3-1 and allow packets in the corresponding VLANs to pass through.

```
[AC]interface GigabitEthernet 0/0/1
[AC-GigabitEthernet0/0/1]port link-type trunk
[AC-GigabitEthernet0/0/1]port trunk allow-pass vlan 20 60
[AC-GigabitEthernet0/0/1]quit
```

7. Configure IP addresses on SW1.

Configure IP addresses and routes on the devices to ensure network connectivity. Create VLANIF interfaces on SW1 and configure IP addresses for the interfaces.

```
[SW1]interface Vlanif 10
```



```
[SW1-Vlanif10]ip address 10.1.10.1 24
[SW1-Vlanif10]quit
[SW1]interface Vlanif 20
[SW1-Vlanif20]ip address 10.1.20.1 24
[SW1-Vlanif20]quit
[SW1]interface Vlanif 30
[SW1-Vlanif30]ip address 10.1.30.1 24
[SW1-Vlanif30]quit
[SW1]interface Vlanif 50
[SW1-Vlanif50]ip address 10.1.50.2 24
[SW1-Vlanif50]quit
[SW1]interface Vlanif 60
[SW1-Vlanif60]ip address 10.1.60.2 24
[SW1-Vlanif60]quit
[SW1]
[SW1]
[SW1]ip route-static 10.10.10.10 32 10.1.60.1
```

8. Configure IP addresses on the AC.

On the AC, create a VLANIF interface and loopback 0, and configure IP addresses for the interfaces.

```
[AC]interface Vlanif 60
[AC-Vlanif60]ip address 10.1.60.1 24
[AC-Vlanif60]quit
[AC]interface LoopBack 0
[AC-LoopBack0]ip address 10.10.10.10 32
[AC-LoopBack0]quit
[AC]
[AC]ip route-static 10.1.10.0 24 10.1.60.2
```

9. Configure an IP address and a static route on R1.

On R1, configure the IP address for the interface connecting to SW1 and a static route to the STA network segment.

```
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.1.50.1 24
[R1-GigabitEthernet0/0/1]quit
[R1]ip route-static 10.1.20.0 24 10.1.50.2
[R1]ip route-static 10.1.30.0 24 10.1.50.2
```

On SW1, ping R1 using the IP address 10.1.20.1. The ping test is successful.

```
[SW1]ping -a 10.1.20.1 10.1.50.1
  PING 10.1.50.1: 56 data bytes, press CTRL_C to break
    Reply from 10.1.50.1: bytes=56 Sequence=1 ttl=255 time=90 ms
    Reply from 10.1.50.1: bytes=56 Sequence=2 ttl=255 time=50 ms
    Reply from 10.1.50.1: bytes=56 Sequence=3 ttl=255 time=40 ms
    Reply from 10.1.50.1: bytes=56 Sequence=4 ttl=255 time=30 ms
    Reply from 10.1.50.1: bytes=56 Sequence=5 ttl=255 time=20 ms

  --- 10.1.50.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
```



0.00% packet loss
round-trip min/avg/max = 20/46/90 ms

3.3 Configuring AP Onboarding

Configure a DHCP server to ensure that STAs and APs can obtain IP addresses.

1. Create an address pool for APs on SW1.

Enable DHCP on SW1 and create an address pool for APs.

```
[SW1]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[SW1]ip pool ap
Info:It's successful to create an IP address pool.
[SW1-ip-pool-ap]network 10.1.10.0 mask 24
[SW1-ip-pool-ap]gateway-list 10.1.10.1
[SW1-ip-pool-ap]option 43 sub-option 2 ip-address 10.10.10.10
[SW1-ip-pool-ap]quit
[SW1]
```

2. Create address pools for employees and guests on SW1.

```
[SW1]ip pool employee
Info:It's successful to create an IP address pool.
[SW1-ip-pool-employee]network 10.1.20.0 mask 24
[SW1-ip-pool-employee]gateway-list 10.1.20.1
[SW1-ip-pool-employee]dns-list 114.114.114.114
[SW1-ip-pool-employee]quit
[SW1]
```

```
[SW1]ip pool guest
Info:It's successful to create an IP address pool.
[SW1-ip-pool-guest]network 10.1.30.0 mask 24
[SW1-ip-pool-guest]gateway-list 10.1.30.1
[SW1-ip-pool-guest]dns-list 114.114.114.114
[SW1-ip-pool-guest]quit
[SW1]
```

3. Enable the global address pool function on interfaces.

```
[SW1]interface Vlanif 10
[SW1-Vlanif10]dhcp select global
[SW1-Vlanif10]quit
[SW1]interface Vlanif 20
[SW1-Vlanif20]dhcp select global
[SW1-Vlanif20]quit
[SW1]interface Vlanif 30
[SW1-Vlanif30]dhcp select global
[SW1-Vlanif30]quit
[SW1]
```

4. Configure the AC so that APs can go online.

Configure the IP address for the AC source interface, and select a proper AP authentication mode to enable APs to go online.

- Run the following command on AP1 to obtain the MAC address of AP1:



```
<Huawei>display system-information
System Information
=====
Serial Number           : 210235448310630DF380
System Time             : 2025-04-06 16:57:02
System Up time          : 28min 20sec
System Name             : Huawei
Country Code            : US
MAC Address            : 00:e0:fc:89:66:e0
Radio 0 MAC Address     : 00:00:00:00:00:00
Radio 1 MAC Address     : 00:00:00:00:00:10
IP Address              : 10.1.10.254
Subnet Mask             : 255.255.255.0
Default Gateway         : 0.0.0.0
IPv6 IP Address         :
IPv6 Default Gateway    :
Management VLAN ID(AP) :
IP MODE                 : dhcp
Slot Status             : Dual band(802.11b/g/n;802.11a/n/ac)
AP Type                 : AP6050DN
Board Type              : AP6050DN
Board Serial Number     : 210235448310630DF380
Board Bom Version       : 0
Boot Rom Version        : -
Software Version        : V200R007C10SPC300
Hardware Version        : H86D2TT1D502 VER.A
Telnet Access           : Enable
User Name               : admin
LED Switch              : ON
=====
<Huawei>
```

- Run the following command on AP2 to obtain the MAC address of AP2:

```
<Huawei>display system-information
System Information
=====
Serial Number           : 21023544831019104516
System Time             : 2025-04-06 16:59:53
System Up time          : 31min 2sec
System Name             : Huawei
Country Code            : US
MAC Address            : 00:e0:fc:49:3c:30
Radio 0 MAC Address     : 00:00:00:00:00:00
Radio 1 MAC Address     : 00:00:00:00:00:10
IP Address              : 10.1.10.253
Subnet Mask             : 255.255.255.0
Default Gateway         : 0.0.0.0
IPv6 IP Address         :
IPv6 Default Gateway    :
Management VLAN ID(AP) :
IP MODE                 : dhcp
Slot Status             : Dual band(802.11b/g/n;802.11a/n/ac)
```



```
AP Type           : AP6050DN
Board Type        : AP6050DN
Board Serial Number : 21023544831019104516
Board Bom Version  : 0
Boot Rom Version   : -
Software Version   : V200R007C10SPC300
Hardware Version   : H86D2TT1D502 VER.A
Telnet Access      : Enable
User Name         : admin
LED Switch        : ON
=====
<Huawei>
```

- Configure loopback 0 as the AC source interface.

```
[AC]capwap source interface LoopBack 0
[AC]
```

- Create the AP group **Huawei** on the AC.

```
[AC]wlan
[AC-wlan-view]
[AC-wlan-view]ap-group name Huawei
Info: This operation may take a few seconds. Please wait for a moment.done.
[AC-wlan-ap-group-Huawei]
[AC-wlan-ap-group-Huawei]quit
[AC-wlan-view]
```

- Set the AP authentication mode to MAC address authentication, and manually add APs to the AP group. Name the APs **AP1** and **AP2**.

AP1's MAC address is **00:e0:fc:89:66:e0**, which needs to be written as **00e0-fc89-66e0**.

AP2's MAC address is **00:e0:fc:49:3c:30**, which needs to be written as **00e0-fc49-3c30**.

```
[AC]wlan
[AC]wlan
[AC-wlan-view]ap auth-mode mac-auth
[AC-wlan-view]ap-mac 00e0-fc89-66e0
[AC-wlan-ap-0]ap-name AP1
[AC-wlan-ap-0]ap-group Huawei
Warning: This operation may cause AP reset. If the country code changes, it will
clear channel, power and antenna gain configurations of the radio, Whether to c
ontinue? [Y/N]:y
Info: This operation may take a few seconds. Please wait for a moment.. done.
[AC-wlan-ap-0]quit
[AC-wlan-view]
[AC-wlan-view]
[AC-wlan-view]ap-mac 00e0-fc49-3c30
[AC-wlan-ap-1]ap-name AP2
[AC-wlan-ap-1]ap-group Huawei
Warning: This operation may cause AP reset. If the country code changes, it will
clear channel, power and antenna gain configurations of the radio, Whether to c
ontinue? [Y/N]:y
Info: This operation may take a few seconds. Please wait for a moment.. done.
[AC-wlan-ap-1]quit
[AC-wlan-view]
```



- Run the **display ap all** command to check the AP onboarding status. The command output shows that the two APs are not online.

```
[AC-wlan-view]display ap all
```

```
Info: This operation may take a few seconds. Please wait for a moment.done.
```

```
Total AP information:
```

```
idle : idle          [2]
```

ID	MAC	Name	Group	IP	Type	State	STA	Uptime
0	00e0-fc89-66e0	AP1	Huawei	-	-	idle	0	-
1	00e0-fc49-3c30	AP2	Huawei	-	-	idle	0	-

```
Total: 2
```

```
[AC-wlan-view]
```

- Troubleshoot the AP onboarding failure.

The troubleshooting result shows that the APs cannot communicate with the AC source address. On SW1, create a static route destined for the AC source address. On the AC, create a static route destined for the AP network segment.

```
[SW1]ip route-static 10.10.10.10 32 10.1.60.1
```

```
[SW1]
```

```
[AC]ip route-static 10.1.10.0 24 10.1.60.2
```

```
[AC]
```

Wait for a while and run the **display ap all** command to check the AP online status. The command output shows that the two APs are online.

```
<AC>display ap all
```

```
Info: This operation may take a few seconds. Please wait for a moment.done.
```

```
Total AP information:
```

```
nor : normal          [2]
```

ID	MAC	Name	Group	IP	Type	State	STA	Uptime
0	00e0-fc89-66e0	AP1	Huawei	10.1.10.254	AP6050DN	nor	0	21S
1	00e0-fc49-3c30	AP2	Huawei	10.1.10.253	AP6050DN	nor	0	11S

```
Total: 2
```

```
<AC>
```

3.4 Configuring WLAN Services

Configure SSID profiles, security profiles, and VAP profiles based on the WLAN data plan to ensure that APs can properly broadcast signals and STAs can connect to the APs.

- Create SSID profiles on the AC.



Create the SSID profiles **Employee** and **Guest**, and set the SSIDs in the profiles to **Employee** and **Guest**, respectively.

```
[AC-wlan-view]ssid-profile name Employee
[AC-wlan-ssid-prof-Employee]ssid Employee
[AC-wlan-ssid-prof-Employee]quit
[AC-wlan-view]
[AC-wlan-view]ssid-profile name Guest
[AC-wlan-ssid-prof-Guest]ssid Guest
[AC-wlan-ssid-prof-Guest]quit
[AC-wlan-view]
```

2. Create security profiles.

Create security profiles **Employee** and **Guest**, and configure security policies.

```
[AC-wlan-view]security-profile name Employee
[AC-wlan-sec-prof-Employee]security wpa2 psk pass-phrase a1234567 aes
[AC-wlan-sec-prof-Employee]quit
[AC-wlan-view]
[AC-wlan-view]security-profile name Guest
[AC-wlan-sec-prof-Guest]security open
[AC-wlan-sec-prof-Guest]quit
[AC-wlan-view]
```

3. Create VAP profiles.

Create the VAP profiles **Employee** and **Guest**, and configure them according to the WLAN data plan.

```
[AC-wlan-view]vap-profile name Employee
[AC-wlan-vap-prof-Employee]ssid-profile Employee
[AC-wlan-vap-prof-Employee]security-profile Employee
[AC-wlan-vap-prof-Employee]service-vlan vlan-id 20
[AC-wlan-vap-prof-Employee]forward-mode tunnel
[AC-wlan-vap-prof-Employee]quit
[AC-wlan-view]
[AC-wlan-view]vap-profile name Guest
[AC-wlan-vap-prof-Guest]ssid-profile Guest
[AC-wlan-vap-prof-Guest]security-profile Guest
[AC-wlan-vap-prof-Guest]service-vlan vlan-id 30
[AC-wlan-vap-prof-Guest]forward-mode direct-forward
[AC-wlan-vap-prof-Guest]quit
[AC-wlan-view]
```

4. Create a regulatory domain profile.

Create the regulatory domain profile **Huawei** and set the country code to **CN**.

```
[AC-wlan-view]regulatory-domain-profile name Huawei
[AC-wlan-regulate-domain-Huawei]country-code CN
Info: The current country code is same with the input country code.
[AC-wlan-regulate-domain-Huawei]quit
[AC-wlan-view]
```

5. Bind profiles.



Enter the view of the AP group **Huawei**, and bind the regulatory domain profile **Huawei**, VAP profile **Employee**, and VAP profile **Guest** to the AP group.

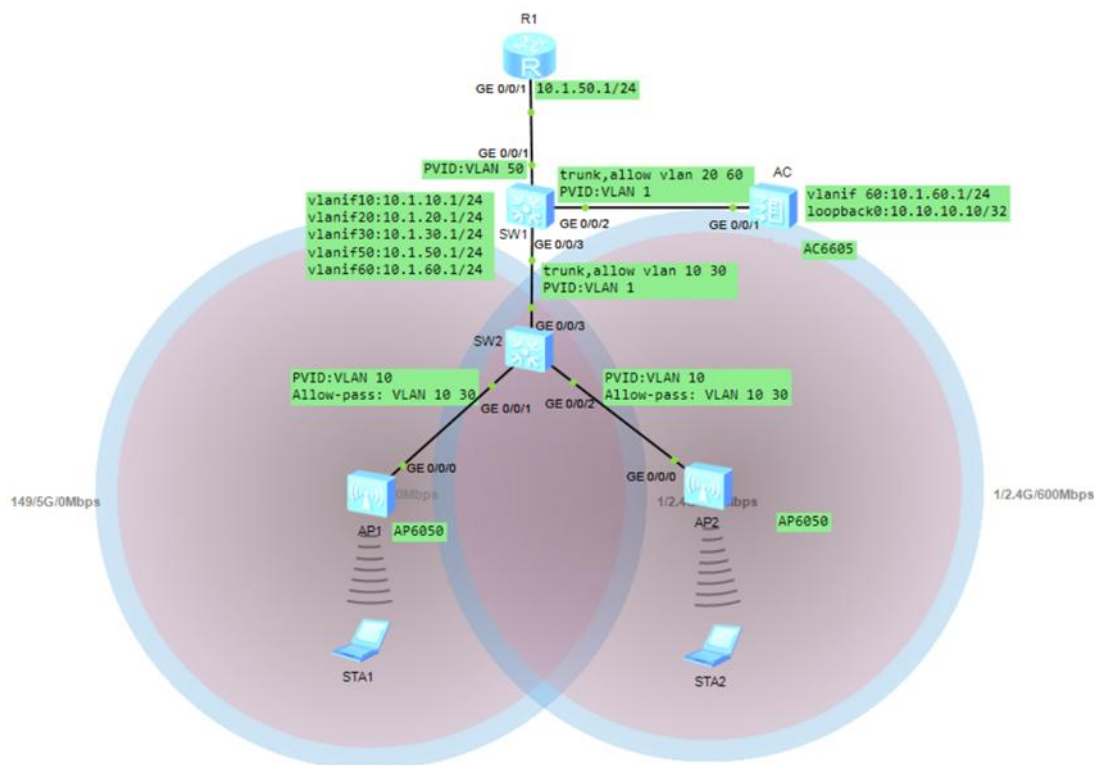
```
[AC-wlan-view]
[AC-wlan-view]ap-group name Huawei
[AC-wlan-ap-group-Huawei]regulatory-domain-profile Huawei
Warning: Modifying the country code will clear channel, power and antenna gain c
onfigurations of the radio and reset the AP. Continue?[Y/N]:y
[AC-wlan-ap-group-Huawei]vap-profile Employee wlan 1 radio all
Info: This operation may take a few seconds, please wait...done.
[AC-wlan-ap-group-Huawei]
[AC-wlan-ap-group-Huawei]vap-profile Guest wlan 2 radio all
Info: This operation may take a few seconds, please wait...done.
[AC-wlan-ap-group-Huawei]
[AC-wlan-ap-group-Huawei]quit
[AC-wlan-view]
```

4 Verification

4.1 Verifying that APs Properly Broadcast Signals

The APs broadcast signals properly, as shown in Figure 4-1.

Figure 4-1 APs broadcasting signals



Run the **display vap ssid Employee** command on the AC to check whether the two APs broadcast signals. The command output shows that the two APs broadcast signals on two radios.



```
[AC]display vap ssid Employee
```

```
Info: This operation may take a few seconds, please wait.
```

```
WID : WLAN ID
```

AP ID	AP name	RfID	WID	BSSID	Status	Auth type	STA	SSID
0	AP1	0	1	00E0-FC89-66E0	ON	WPA2-PSK	0	Employee
0	AP1	1	1	00E0-FC89-66F0	ON	WPA2-PSK	0	Employee
1	AP2	0	1	00E0-FC49-3C30	ON	WPA2-PSK	0	Employee
1	AP2	1	1	00E0-FC49-3C40	ON	WPA2-PSK	0	Employee

```
Total: 4
```

```
[AC]
```

```
[AC]display vap ssid Guest
```

```
Info: This operation may take a few seconds, please wait.
```

```
WID : WLAN ID
```

AP ID	AP name	RfID	WID	BSSID	Status	Auth type	STA	SSID
0	AP1	0	2	00E0-FC89-66E1	ON	Open	0	Guest
0	AP1	1	2	00E0-FC89-66F1	ON	Open	0	Guest
1	AP2	0	2	00E0-FC49-3C31	ON	Open	0	Guest
1	AP2	1	2	00E0-FC49-3C41	ON	Open	0	Guest

```
Total: 4
```

```
[AC]
```

4.2 Verifying STA Connections and Network Connectivity

Connect STAs to the two signals of APs and ping R1 to test network connectivity.

- SSID: Guest

```
STA>ipconfig
```

```
Link local IPv6 address.....: ::  
IPv6 address.....: :: / 128  
IPv6 gateway.....: ::  
IPv4 address.....: 10.1.30.254  
Subnet mask.....: 255.255.255.0  
Gateway.....: 10.1.30.1  
Physical address.....: 54-89-98-73-77-47  
DNS server.....: 114.114.114.114
```

```
STA>ping 10.1.50.1 -c 2
```

```
Ping 10.1.50.1: 32 data bytes, Press Ctrl_C to break  
From 10.1.50.1: bytes=32 seq=1 ttl=254 time=187 ms  
From 10.1.50.1: bytes=32 seq=2 ttl=254 time=157 ms
```

```
--- 10.1.50.1 ping statistics ---  
2 packet(s) transmitted
```



```
2 packet(s) received
0.00% packet loss
round-trip min/avg/max = 157/172/187 ms
```

STA>

- SSID: Employee (with the password of **a1234567**)

STA>ipconfig

```
Link local IPv6 address.....: ::
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.1.20.254
Subnet mask.....: 255.255.255.0
Gateway.....: 10.1.20.1
Physical address.....: 54-89-98-A4-36-26
DNS server.....: 114.114.114.114
```

STA>ping 10.1.50.1 -c 2

```
Ping 10.1.50.1: 32 data bytes, Press Ctrl_C to break
From 10.1.50.1: bytes=32 seq=1 ttl=254 time=187 ms
From 10.1.50.1: bytes=32 seq=2 ttl=254 time=218 ms
```

```
--- 10.1.50.1 ping statistics ---
2 packet(s) transmitted
2 packet(s) received
0.00% packet loss
round-trip min/avg/max = 187/202/218 ms
```

STA>

5 Quiz

1. What are the advantages of in-path and off-path networking modes?

In-path networking advantages: Direct forwarding is often used. This networking mode simplifies the network architecture and applies to large-scale centralized WLANs.

Off-path networking advantages: The off-path networking mode is commonly used. STA service data does not need to be processed by an AC, eliminating the bandwidth bottleneck and facilitating the usage of existing security policies. Therefore, this networking mode is recommended.

2. What are the advantages and disadvantages of tunnel forwarding?

Tunnel forwarding mode:

Advantages: An AC forwards all data packets, ensuring security and facilitating centralized management and control.

Disadvantages: Service data must be forwarded by an AC, which is inefficient and increases the load on the AC.

3. What are the advantages and disadvantages of direct forwarding?



Direct forwarding mode:

Advantages: Service data packets do not need to be forwarded by an AC, improving packet forwarding efficiency and reducing the burden on the AC.

Disadvantages: Service data is difficult to manage and control in a centralized manner.